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Investigating high school students' perceptions and presences under VR learning environment

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ABSTRACT
Recently, Virtual Reality (VR) has developed rapidly. Some studies propose that VR has a great deal of potential as an instructional media for learning, while other studies indicate that VR may be ineffective or even negatively influence learners' learning experience. Few studies provide reference to educators or teachers based on empirical data. This study investigates 55 10th-grade learners' perceptions and presences in a VR-based blended learning environment. A research framework based on social presence theory and the Community of Inquiry model (CoI) was adopted in order to study social presence, teaching presence, and cognitive presence of learners in a VR-based practical, blended learning environment. The research findings are as follows: (1) VR enhanced learners' perception of the teaching equipment and the classroom, social presence, teaching presence. (2) VR-based blended learning can have a positive effect on the interest, attention, and interaction of learners. (3) For transferring well-structured knowledge, there is no significant difference between traditional instructional media and VR. But for transferring ill-structured knowledge, VR is more effective at promoting learning than traditional instructional media. In general, VR-based learning environments can positively affect students' perceptions and presence.

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VR; perception; presence; CoI; learning experience

Introduction

Education is a collaborative reconstruction of learning experience (Dewey, 1897). Effective learning is inseparable from well-defined instructional contexts and well-designed digital learning environments (Dillenbourg & Hong, 2008). Due to innovations in science and technology, great changes have taken place in teaching and learning environments. As increased numbers of people have opportunities to improve their learning performance by using various computer media (OECD, 2015; Rashid, & Asghar, 2016), they become extensive users of information and communication technologies (ICT) (Horrigan, 2007).

At present, an emerging technology, virtual reality (VR) is attracting wide attention in the field of education because of its potential to assist learning in different ways. Since the early 1990s, VR technology has been used in K-12 and higher education (Youngblut, 1998). Gradually, it has been proven that the application of VR is an interesting and promising area in education (Ma & Mak, 2000).

Although some studies find that VR is greatly helpful to promote teaching (Alfalah, 2018; Falah et al., 2014; Mano, 2019), other studies find that effects of using VR are still limited or have yet to

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be verified (Moro, Stromberga, & Stirling, 2017; Wang & Lindeman, 2015). Technical difficulties and practical uses are the two main problems in VR-based blended learning at the moment. First, various kinds of technical difficulties make it difficult to meet the high-resolution requirements needed to transmit VR content, breaking users' immersive experience and even causing health and other discomfort problems, such as cybersickness (Domingo & Bradley, 2018; Elbamby, Perfecto, Bennis, & Doppler, 2018; Huang et al., 2019; Jensen & Konradsen, 2018). Second, in some cases, current VR technology fails to exactly identify facial expressions and senses and accurately track eye points (Bastug, Bennis, Médard, & Debbah, 2017); furthermore, users' vision can be obstructed. These difficulties cause practical application obstacles in VR-based learning environments.

Overcoming or mitigating these difficulties is important for VR-based learning. Most studies show that the educational use of VR needs a paradigm shift (Jensen & Konradsen, 2018; Liu, Bhagat, Gao, Chang, & Huang, 2017), or it will likely have a negative impact on learners' learning experience (Chen, Chao, Chen, Wang, & Tan, 2015; Hashemian & Riecke, 2017; Pellas & Boumpa, 2017). There is currently insufficient empirical evidence of VR's educational value or existing problems; there is also no clear guidance on how to better use VR to assist instruction (Makransky, Terkildsen, & Mayer, 2017). For example, although previous studies proved virtual learning environments help construct learning experiences, few studies have investigated the influence of social presence in learning and teaching in virtual worlds or in a blended learning environments (Cho, Yim, & Paik, 2015).

The purpose of the present study is to explore how students view the teacher's social presence of using VR-based learning through empirical methods on a practical case. The present study uses Community of Inquiry (CoI), a classic research model. The following questions guided the analysis and are addressed in this study:

(1) Does VR have an impact on learners' perceptions and presences? If so, is this effect positive or negative?
(2) Is there any difference in impact if VR is applied to different lessons?
(3) What are the students' views on introducing VR into the classroom?

Literature review

VR allows users to experience an electronic simulated environment. Early on in VR development, users interacted with the virtual environment (VE) mainly via head-mounted eye goggles and wired clothing (Coates, 1992). Through technological innovation, wearable VR devices have become smaller and more portable. Nowadays, a VR device is regarded as a collection of hardware and software, including a personal computer (PC), head-mounted displays (HDMs), tracking sensors and other components, plus software to deliver an immersive experience (Hussein & Nätterdal, 2015). Because most VR user perceptions rely on computer-generated context, VR also can be regarded as an illusion generated in the human brain. VR immerses people in a relatively independent space (Chu, Chen, Tan, & Zheng, 2019).

VR has been applied in undergraduate courses like medical anatomy (Falah et al., 2014), spacecraft training (Aoki, Oman, & Natapoff, 2007), and architectural simulation (Sampaio, Ferreira, Rosário, & Martins, 2010). When VR is introduced into teaching as a teaching media, the main users—students—should consider it useful and regard it as a valuable supplementary tool in the learning process (Alfalah, 2018). By using VR devices, abstract knowledge can be visualized and presented to learners with 3D views. In such an interactive, fascinating, and immersive situation, learners are more likely to have a pleasant and realistic learning experience (Alfalah, 2018).

Recently, many educators have begun to pay attention to the application of VR in teaching (Lameras, Savin-Baden, Petridis, Dunwell, & Liarokapis, 2014; Muir, Allen, Rayner, & Cleland, 2013; Zhao, Chis, Muntean, & Muntean, 2018), or integrated VR technology into their daily teaching to

explore the effects of the VR environment (Erbay, Şimşek, & Kirişçi, 2019; Ghergulescu et al., 2018; Guerrero, Ayala, Mateu, Casades, & Alamán, 2016; Merchant, Goetz, Cifuentes, Keeney-Kennicutt, & Davis, 2014). In addition, some researchers have focused on teachers' attitudes toward using VR (Chen, Shih, & Yu, 2012; Keskitalo, 2011; Sáez-López, Miller, Vázquez-Cano, & Domínguez-Garrido, 2015) and students' perceptions and evaluations for VR (Bogusevschi, Muntean, & Muntean, 2019; Huang, Rauch, & Liaw, 2010; Pantelidis, 2010; Simsek, 2016), to verify the attitude about, and acceptance of, teachers and students using VR in teaching.

However, VR is hindered by many practical problems, which has limited its wide application in education. Whether VR is suitable for use in real classroom situations, there is currently insufficient empirical evidence to verify its educational value (Makransky et al., 2017). Some studies show that when learners spend a long time in an immersive space isolated from the physical environment, they may be more prone to motion sickness or disorientation (Chen et al., 2015). In addition, some learners who have not experienced a 3D virtual environment before may experience cognitive overload while exploring inside one (Pellas & Boumpa, 2017). Therefore, although VR shows promise in the educational realm, it is not certain that introducing VR there would have a positive impact on students' learning experience.

Using empirical methods, this study investigated students' real perceptions and presences for VR in the classroom in order to identify the effectiveness of VR on students' learning experience. In terms of how students view teacher's social presence for VR, Simsek (2016) explored the change in students by using Second Life to present concrete 3D objects in mathematics courses and revealed that using VR equipment in the learning process dramatically affected students' attitude toward mathematics in a positive way. Besides, Bogusevschi et al. (2019) designed and developed a 3D immersive physics educational application to assist the teacher to teach the water cycle in nature, finding that the students in the experimental group preferred to take part in the lessons with such a novel approach. Other studies have explored the effect of different levels of authenticity, immersion, and emotion on learning retention and students' relationships with each other (Reiners et al., 2014).

Gunawardena's (1995) social presence theory and Garrison's (2000) CoI model provide a useful theoretical framework for the present study. The questionnaire was developed based on Gunawardena's social presence items and the CoI model in order to explore the views of VR in secondary education from the aspects of cognitive presence, teaching presence, social presence, and students' feedback. Cognitive presence, teaching presence, and social presence are components of the CoI model, so it's appropriate that the CoI model was used to design the research.

Theoretical framework

Social presence analysis

Gunawardena (1995) and Tu and Mcisaac (2002) define social presence as a perception in person-to-person communication; social presence can also be structured and maintained by human behavior. The study of social presence was inspired by the common concern of some communication researchers that the spread of computer-mediated communication (CMC) might lead to the loss of learners' sense of belonging to other students, teachers, learning projects and teaching institutions (Wiener & Mehrabian, 1968). The term "social presence" was initially put forward by Short, Williams, and Christie (1976); it was defined as the degree to which a person is perceived as a "real person" and connected to others in the process of communication through a medium. Since the 1990s, due to the popularization of computer and Internet, online learning has developed rapidly. Some researchers attempt to use the theory of social presence to explain the dehumanization characteristics of CMC. One of the most important reasons why virtual learning environments are criticized is the lack of social clues, which can create a sense of isolation and separateness and is likely to have a negative impact on teachers' teaching and students' learning (Berge & Collins, 1995).

The intrinsic attribution and factors of social presence have always been bones of contention. Researchers fall into two camps: technology and social contact. Short et al. (1976) reckon that social presence is an attribution of medium, and since the interaction between interlocutors at both ends of the medium need not be considered, different medium environments lead to different social presences. Other adherents of the technology camp assert that the main factors that influence social presence can be divided into interactive responses, affective response, and cohesive responses (Rourke, Anderson, Garrison, & Archer, 2001). However, adherents of the social contact camp maintain that factors influencing social presence include not only social context, online communication, interactivity, system privacy, and feeling of privacy (Tu & Mcisaac, 2002), but also learners' sociability and group behavior in social space (Kreijins, Kirschner, Jochems, & Buuren, 2011).

Recently, more studies of social presence have been conducted in the field of online learning, distance education, and virtual learning environments (VLEs). Teachers with high social presence are more active and effective, and their better teaching results in better course teaching (Grieve, Padgett, & Moffitt, 2016; Hostetter & Busch, 2013). Similarly, learner's ideal learning environment cannot be achieved without the support of social presence (Stacey, 2002). To a large extent, social presence dominates the level of interaction, learning enthusiasm and learning result of the learner in the learning process (Teng, 2013). Further, social presence can also be a predictive indicator of learning satisfaction (So & Brush, 2008) and instructor satisfaction (Richardson & Swan, 2003). These findings indicate that social presence plays an important role in learning processes and outcomes (Cho et al., 2015).

Community of inquiry (CoI) model

While studies of social presence predate the establishment of the CoI model in 2000 by 20 years, Garrison, Anderson, and Archer (2000) put forward a comprehensive CoI model consisting of social presence, teaching presence, and cognitive presence. In the CoI model, as shown in Figure 1, learning experiences in community are regarded as a function of the relationship of the three presences and the interactions among them (Garrison et al., 2000; Ice, 2008; Pellas & Kazanidis, 2014). In recent years, researchers have used CoI models to explore varied learning experiences in different learning environments, such as blended learning environments (Chang-Tik, 2017; Pool, Reitsma, &

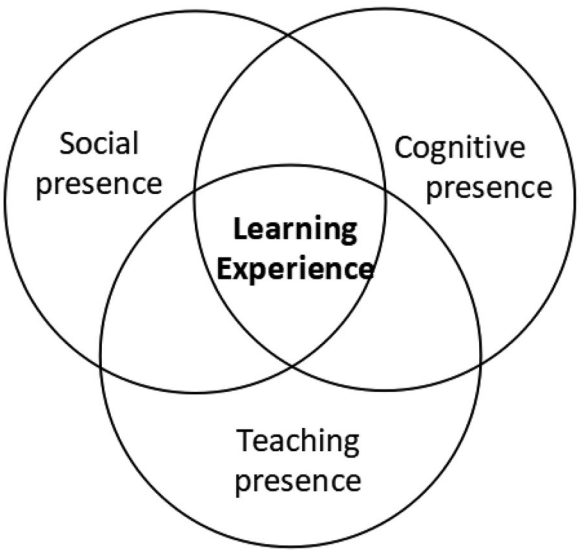

Figure 1. CoI framework.

Van den Berg, 2017; Zafar, Khan, & Hussain, 2017; Zulkanain & Rahim, 2017a, 2017b) and online learning environments, such as Facebook, Moodle, or MOOCs (Annamalai, Tan, & Abdullah, 2016; Kaul, Aksela, & Wu, 2018; Kazanidis, Pellas, Fotaris, & Tsinakos, 2018; Keles, 2018; Padilla & Kreider, 2018; Parker, 2017; Xu, Luo, & Tan, 2018). However, studies of VR-based blended environments in classroom practice have yet to be conducted under the CoI framework.

In the CoI model, teaching presence is described as the design, facilitation and direction of cognitive and social processed that can promote learner's meaningful and effective learning (Chang-Tik, 2017; Garrison et al., 2000). In VLEs, interaction among students is essential; however, interaction cannot guarantee effective learning (Annamalai et al., 2016; Kaul et al., 2018; Xu et al., 2018). Further, the sense of the teacher's presence influences students' satisfaction more than peers' presence (Swan & Shih, 2005); therefore, it's important to put forward teaching presence in order to ensure the learning parameters and specific directions. Later researchers simplified teaching presence into two components: instructional design and direct facilitation (Shea, Pickett, & Pelz, 2003). Instructional design is regarded as the learner's perception of the courses as organized and instructed by teachers according to the course purpose, while direct facilitation is regarded as learner's perception of teacher's encouragement and feedback (Jung & Lee, 2018). In course teaching, activities such as creating slides, developing audio and video, and arranging group discussion are particularly important. Because a clear and consistent curriculum structure can provide strong support for busy teachers and dynamic course discussions, teaching presence has become one of the most important predictors of teaching success (Swan, 2002), learner's engagement (Hew, 2016) and learning persistence (Adamopoulous, 2013; Chang, Hung, & Lin, 2015).

Compared with social presence and teaching presence, the initial definition of cognitive presence has been relatively stable and has the highest acceptance. Initially defined as a practical inquiry model, this model of cognitive presence is composed of four parts, including triggering events, exploration, integration, and resolution (Garrison et al., 2000). Subsequently, Garrison et al. (2000) refined it into the following definition: "*Cognitive presence is the degree to which the learner constructs and confirms meaning through continuous reflection and discourse*". Furthermore, social presence and teaching presence can affect cognitive presence (Garrison, 2007). Therefore, cognitive presence also reflects the process of constructing collaborative knowledge based on learning experiences in constructivism. Although under the CoI model, there are fewer studies of cognitive presence than those of teaching presence and social presence, cognitive presence is the most important core element to explore in learning communities of practice (Swan et al., 2008).

The measuring tools of cognitive presence are limited; most questionnaires and scales that can simultaneously measure teaching presence, social presence, and cognitive presence are adapted from an instrument proposed by Garrison, Cleveland-Innes, and Fung (2004). Using the CoI model to measure cognitive presence may be limited, but this model has successfully measured the likelihood of the development of interdisciplinary learning processes in technology contexts (Arbaugh et al., 2008). According to Arbaugh et al. (2008), using CoI as an independent measure to compare the implementation of curriculum and emerging technologies within the curriculum should be encouraged.

Materials and methods

Our empirical experiment focused on two lessons with different content during a two-week period in March, 2019. It took place in a classroom with integrated information technology and curriculum. The participants were students from two classes and their teacher. The study aimed to examine the presences and perceptions of students in terms of using VR in an authentic class and explore the practical effects of using VR in teaching. To do so, we adopted the traditional COI framework as our theoretical basis and developed an open-ended questionnaire by integrating three existing questionnaires. After finishing the questionnaire, the participants were interviewed based on the following research questions: (a) Does virtual reality have any influence on learners' perception and existence?

If so, is the effect positive or negative? (b) Does VR have different influences when it is applied to different learning content? (c) How do students feel about VR being introduced into the classroom?

Participants

A total of 55 juniors in a high school in Beijing, China were selected for this study; they were between 13 and 15 years old. These participants came from two classes: 27 from class A and 28 from class B. The participants were recruited because they were observed properly operating the VR equipment to some extent. The participants from each of the two classes were asked to complete a pretest questionnaire to ensure that the students in both classes were similar with respect to VR use, social interaction, and cooperative learning. With their similarity across classes established, the participants from two classes separately engaged two lessons, with one lesson implemented using immersive VR, and the other lesson implemented using an overhead projector (OHP). Each lesson was taught by the same teacher. After finishing each lesson, the participants from two classes were asked to answer the main questionnaire (Q) to explore the difference in authentic teaching between VR use and OHP use.

Research design

In this study, both class A and class B were taught two lessons by the teacher A in subject integration courses. The contents of these two lessons were independent; the first lesson was an integration of VR and architecture focusing on appreciation of classical structures, while the second lesson was an integration of VR and biology on characteristics of the nervous system. VR devices were used to construct a VLE and present 3D learning scenes for students. In order to reduce the impact of learning content and VR use order, the research design is shown in Figure 2. Before the experiment, pretest questionnaires (hereafter referred to as pretests) were distributed to the students of the two classes to determine whether there was a difference in the two classes in terms of social presence. After each lesson, students were asked to fill out the main questionnaire (Q). This study analyzed learning differences using VR with two different lessons (different learning content) between two classes, including differences in the first lesson, differences in the second lesson, comprehensive analysis of the two lessons, as well as students' feedback and suggestions for VR.

The OHP condition (control condition)

The OHP setting was a traditional multimedia class for lecture where the participants used an electronic whiteboard to post PowerPoint slides, pictures, and animation videos related to the teaching content. Participants learned the content mainly through teachers' demonstrations via pictures and videos.

VR equipment

The VR equipment used in this study was a 3D virtual simulation laboratory based on G-Motion (Manheng Digital Technology, Shanghai, China, http://www.gdi.com.cn). G-Motion is composed of many parts, including camera, image workstation, speaker system, motion capture and position tracking system, and so on. The pose and motion information of head, hand, and other key parts of the body can be captured in real-time. In addition, G-Motion can capture 6-DOF (positions and direction) information of the target object in real-time for gait analysis and virtual human motion simulation.

In this study, students used VR glasses or hand-held interactive devices to interact in the virtual environment to focus on observing objects from any angle. Tracking accuracy is up to 0.2 mm and angle tracking accuracy is up to 0.2 degree, so the precise interaction of human and virtual environment could be noted. For example, in the first lesson, students could pause and zoom in on the interesting parts of the structure of an ancient building, leading to a more comprehensive understanding of the teaching content. Actual pictures of students using VR devices are shown in Figures 3 and 4.

	Before the experiment	The first lesson（An integration of VR and architecture）		The second lesson（An integration of VR and biology on characteristics of the nervous system）
A Class	Pretest Pre A	Use VR device in class Q_A1		Do not use VR device in class Q_A2
B Class	Pretest Pre B	Do not use VR device in class Q_B1		Use VR device in class Q_B2
Purpose	Analyze Pre_A and Pre_B To determine whether there is a difference in the two classes	Analyze Q_A1 and Q_B1 To compare the effect of VR assisted instruction in the first lesson	Analyze (Q_A1, Q_B2) & (Q_A2, Q_B1) To Comprehensively compare the effect of whether using VR assisted instruction	Analyze Q_A2 and Q_B2 To compare the effect of VR assisted instruction in the second lesson

Figure 2. Study design.

In addition, other tools used in this study included an electronic whiteboard, which was used to post PowerPoint slides, pictures, and animation videos of structures of buildings and nervous systems. Students could understand teaching content mainly through teachers' demonstrations of pictures and videos, meaning in a traditional classroom lecture style. Using both traditional and

Figure 3. Actual pictures of students using VR devices.

Figure 4. Actual pictures of students using VR devices.

VR-assisted approaches in practice, students could all understand all the teaching content of class. The classroom atmosphere was active in both classes.

Instruments and scale validations

There were two questionnaires used in this study: the pretest questionnaire and main questionnaire. Because social presence can be affected by learners' sociability and group behavior in social space (Kreijins et al., 2011), the pretest was used to explore whether there was any significant difference in social presence between Class A and Class B students. The main questionnaire was used to explore the evaluation of the use of VR equipment, and the learning experience of the learners using a VLE in the physical classroom.

The pretest was composed of three dimensions, including perception of the assistance of group activity to learning (w1), social comfort of expressing and sensing affect (w2), and social navigation (w3) respectively. This questionnaire was developed by Lin (2004), based on work by Picciano (2002) and Tu (2001), and its effectiveness has been proven. It was included in the revised scale used in the present study.

The main questionnaire consisted of 47 items in five dimensions, including teaching equipment perception (m), class perception (n), social presence (p), teaching presence (r) and cognitive presence (t). In addition, there were two open-ended questions at the end of questionnaire. The measurements of teaching equipment perception (m) and class perception (n) were based on the study of 3D VLE by Matsukawa and Nishimori (2014). The measurement of the social presence dimension (p) was based on 17 adjectives proposed by Gunawardena (1995); these items are used to describe media characteristics. Finally, taking as a reference Arbaugh's (2008) scale, which is based on the CoI model, the last two dimension measurement items were determined. Because the participants in the present study were high school students, some statements of questionnaires were adapted to be more comprehensible. The two open-ended questions are presented as follows: "Do you like learning with VR

equipment? Why?" and, "Do you have any comments or suggestions for the use of VR equipment in this lesson?"

The scoring rules of these two questionnaires are as follows. The 17 items of social presence are based on a seven-point scale, ranging from "Completely inconsistent" to "Completely consistent" with scores from 1 to 7. In addition to social presence, other dimensions use a Likert scale, ranging from "Completely inconsistent" to "Completely consistent" with scores from 1 to 5. The implications of the score were defined, the higher the score, the more likely a student is to agree with the statement.

A total of 55 pretests were collected, with a recovery rate of 100% and a utilization rate of 90%. A total of 110 main questionnaires were collected; the recovery rate of the main questionnaire was 100%, and the utilization rate was 82.72%. Valid questionnaires include 21 Q_A1 and 26 Q_A2 in Class A, and 21 Q_B1, and 23 Q_B2 in Class B.

Kreijins's (2011) questionnaire was employed in this study. The reliability test was carried out on the three dimensions of the pretest, and the Cronbach's Alpha values of the three dimensions are all greater than 0.8, indicating the high reliability of the pretest. The reliability test result of the pretest is shown in Table 1.

The items of five dimensions of the main questionnaire were based on previous studies, including teaching equipment perception and class perception (Matsukawa & Nishimori, 2014), social presence (Gunawardena, 1995), and teaching presence and cognitive presence (Arbaugh, 2008). The reliability of each dimension of the main questionnaire was tested. It was found that the Cronbach's Alpha values of all dimensions is greater than 0.85. According to the internal consistency reliability evaluation index, the reliability of the questionnaire is ideal, and the items have very high consistency. The reliability test result of the main questionnaire is shown in Table 2.

Data analysis

Social science statistics software (SPSS 22.0) was used in the study to process the data, focusing on using the nonparametric technique, the Mann–Whitney U test. After the participants finished their learning content, the teacher distributed questionnaires to each participant to gather data.

Results

Analysis of pretest questionnaire

Before analyzing how students view teacher's social presence under the experimental and control conditions, a pretest was given to ensure the participants from the two classes were at the same level in two dimensions: cooperation and social competence. The Mann–Whitney U test results found that there was no significant difference in the levels of cooperation (A: $Z = -0.425$, $P = 0.671 > 0.05$;) and social competence (w1: $Z = -0.425$, $P = 0.671 > 0.05$; B: $Z = -1.132$, $P = 0.258 >$

Table 1. Reliability test result of the pretest.

Dimensions	Perception of the assistance of group activity to learning	Social comfort of expressing and sensing affect	Social navigation	All items
Cronbach's Alpha	.909	.818	.829	.931

Table 2. Reliability test result of the main questionnaire.

Dimensions	Teaching equipment perception (m1-m7)	Class perception (n8-n16)	Social presence (p17-p33)	Teaching presence (r34-r42)	Cognitive presence (t43-t47)	All items
Cronbach's Alpha	.872	.901	.962	.920	.870	.972

0.05; C: Z = −0.037, $P = 0.970 > 0.05$) between participants from the two classes, as shown in Table 3. It could be determined that the two classes used in the experiment were parallel classes, and the personal interference factors of the two classes before the experiment can be excluded.

Analysis of differences between the control and experimental conditions in the first lesson

In the first lesson, class A did not use VR, and so it can be considered the control condition, while class B used VR, so it can be considered the experimental condition. The Mann–Whitney U test was conducted on the main questionnaire collected after the first class, and the results are shown in Table 4.

As shown in Table 4, it was found that there was no significant difference between the VR experimental condition and the OHP control condition with respect to teaching equipment perception (m), class perception (n), social presence (p), teaching presence (r), and cognitive presence (t).

Of the 47 items on the main questionnaire, 46 items showed no significant differences between the experimental and control conditions. The one item with a significant difference between the experiment and the control was p17, "exciting," of social presence with sig (2-tailed) value 0.015. Participants reported that learning under the immersive VR experimental condition was more exciting than that under the OHP control condition.

Analysis of differences between the control and experimental conditions in the second lesson

In the second lesson, class A used VR (experimental condition), and class B did not use VR (control condition). The Mann–Whitney U test was conducted to identify if there were differences of presences and perceptions between the two classes. The results are shown in Table 5.

As is shown in Table 5, It is revealed that there are significant differences between the control and experimental condition under the second lesson in teaching equipment perception (M: Z = −3.012, $P = 0.003 < 0.05$), class perception (N: Z = −2.711, $P = 0.007 < 0.05$), teaching presence (R: Z = −2.054, $P = 0.040 < 0.05$), and cognitive presence (T: Z = −2.550, $P = 0.011 < 0.05$). Among the 47, 14 questions (distributed on 5 dimensions) showed significant differences; the scores of these 14 questions are higher under the OHP control condition. Detailed data analysis of these 14 questions are as following in Table 5.

First, in terms of teaching equipment perception (M) in Table 6, there were some significant differences between the VR experimental condition and the OHP condition. Students reported higher scores for VR on items such as "small words are clearly visible" (m1: Z = −2.542, $P = 0.011 < 0.05$), "eyes do not get tired easily" (M2: Z = −2.562,$P = 0.010 < 0.05$), "learning content can be presented clearly" (M5: Z = −2.307, $P = 0.021 < 0.05$), "audio can be heard clearly" (M6: Z = −2.303, $P = 0.021 < 0.05$), and "I feel it looks real" (M7: Z = −3.333, $P = 0.001 < 0.05$). These results show that, for the second lesson, the participants' perception of teaching equipment was higher in the VR experimental condition than that in the OHP control condition.

Second, for classroom perception (N), significant differences existed in four following items: "teacher's speaking style and teaching method are easy to understand" (N12: Z = −2.143, $P = 0.032 < 0.05$), "the presentation of learning content is easy to understand" (N13: Z = −3.006, $P = 0.003 < 0.05$), "I personally feel helped by the teacher" (N14: Z = −2.127, $P = 0.033 < 0.05$), "I can clearly understand the class content" (n15: Z = −2.101, $P = 0.036 < 0.05$). These differences suggest that teachers in VR learning environments are more sensitive to the students' needs than those in the OHP control condition Table 7.

Third, in terms of social presence (P), there were no significant differences as a whole, but by analyzing the data, we found that partial items of social presence still showed significant differences, such as "appealing" (p24: Z = −2.284, $P = 0.022 < 0.05$), "personality" (p28: Z = −2.005, $P = 0.045 < 0.05$), and "non-threatening" (p32: Z = −2.015, $P = 0.044 < 0.05$). The results show that participants

Table 3. Two-Independent-Samples tests of pretest.

Test Statistics[a]															
	a1	a2	a3	a4	a5	A	b6	b7	b8	b9	B	c10	c11	c12	C
Mann-Whitney U	320.500	314.500	287.000	325.000	313.000	315.000	329.500	244.000	285.000	328.500	276.500	320.500	326.000	328.000	336.000
Wilcoxon W	671.500	665.500	638.000	676.000	664.000	666.000	680.500	595.000	636.000	679.500	627.500	671.500	677.000	679.000	687.000
Z	−.348	−.456	−.976	−.264	−.490	−.425	−.161	−1.775	−1.041	−.184	−1.132	−.341	−.238	−.194	−.037
Asymp. Sig.(2-tailed)	.728	.649	.329	.792	.624	.671	.872	.076	.298	.854	.258	.773	.812	.884	.970

[a]Grouping Variable : Class.

Table 4. Two-Independent-Samples tests of the first lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
M	10	21	21.00	441.00
	11	21	22.00	462.00
	Total	42		
N	10	21	21.93	460.50
	11	21	21.07	442.50
	Total	42		
p17	10	21	25.90	544.00
	11	21	17.10	359.00
	Total	42		
P	10	21	24.48	514.00
	11	21	18.52	389.00
	Total	42		
R	10	21	23.64	496.50
	11	21	19.36	406.50
	Total	42		
T	10	21	21.21	445.50
	11	21	21.79	457.50
	Total	42		

Test Statistics[a]						
	M	*N*	P17	*P*	*R*	*T*
Mann-Whitney U	210.000	211.500	128.000	158.000	175.500	214.500
Wilcoxon W	441.000	442.000	359.000	389.000	406.5.000	445.500
Z	−.266	−.228	−2.442	−1.576	−1.138	−.152
Asymp. Sig.(2-tailed)	.791	.820	.015	.115	.255	.879

[a]Grouping Variable : Group.

Table 5. Two-Independent-Samples tests of the second lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
M	20	26	30.71	798.50
	21	23	18.54	426.50
	Total	49		
N	20	26	30.17	784.50
	21	23	19.15	440.50
	Total	49		
P	20	26	28.44	739.50
	21	23	21.11	485.50
	Total	49		
R	20	26	28.87	750.50
	21	23	20.63	474.50
	Total	49		
T	20	26	29.85	776.00
	21	23	19.52	449.00
	Total	49		

Test Statistics [a]					
	M	*N*	*P*	*R*	*T*
Mann-Whitney U	150.500	164.500	209.500	198.500	173.000
Wilcoxon W	426.500	440.500	485.500	474.500	449.000
Z	−3.012	−2.711	−1.794	−2.054	−2.550
Asymp. Sig. (2-tailed)	.003	.007	.073	.040	.011

[a]Grouping Variable : Group.

in the VR experimental condition have more obvious social experience than those under the OHP control condition; this suggests that VR has an influence on social presence for students.

Fourth, for teaching presence (T), students indicated that the answers given by the teacher were more detailed and easier to understand in the VR learning environment (R42: $Z = -2.321$, $P = 0.020 < 0.05$), as shown in Table 8. This result suggests that using VR benefits the teacher by simplifying teaching. Lastly, in terms of cognitive presence, there was significant difference in "I will

Table 6. Two-Independent-Samples tests of teaching equipment perception in the second lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
m1	20	26	29.38	764.00
	21	23	20.04	461.00
	Total	49		
m2	20	26	29.69	772.00
	21	23	19.70	453.00
	Total	49		
m5	20	26	28.81	749.00
	21	23	20.70	476.00
	Total	49		
m6	20	26	28.60	743.50
	21	23	20.93	481.50
	Total	49		
m7	20	26	30.65	797.00
	21	23	18.61	428.00
	Total	49		

Test Statistics[a]					
	m1	m2	m5	m6	m7
Mann-Whitney U	185.000	177.000	200.000	205.500	152.000
Wilcoxon W	461.000	453.000	476.000	481.500	428.000
Z	−2.542	−2.562	−2.307	−2.303	−3.333
Asymp. Sig. (2-tailed)	.011	.010	.021	.021	.001

[a]Grouping Variable : Group.

Table 7. Two-Independent-Samples tests of class perception in the second lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
n12	20	26	28.62	744.00
	21	23	20.91	481.00
	Total	49		
n13	20	26	30.27	787.00
	21	23	19.04	438.00
	Total	49		
n14	20	26	28.38	738.00
	21	23	21.17	487.00
	Total	49		
n15	20	26	28.83	749.50
	21	23	20.67	475.50
	Total	49		

Test Statistics[a]				
	n12	n13	n14	n15
Mann-Whitney U	205.000	162.000	211.000	199.500
Wilcoxon W	481.000	438.000	487.000	475.500
Z	−2.143	−3.006	−2.127	−2.101
Asymp. Sig. (2-tailed)	.032	.003	.033	.036

[a]Grouping Variable : Group.

apply the knowledge learned from this lesson to other activities in the future" (T45: $Z = -2.692$, $P = 0.007 < 0.05$) Table 9.

These results show that the use of VR in the second lesson significantly enhanced class B students' performance in all dimensions, especially in teaching equipment perception, classroom perception, social presence and teaching presence.

Comprehensive analysis of differences between control and experimental conditions in two lessons

The Mann–Whitney U test was used to comprehensively analyze the questionnaires of the participants in the two classes who used VR equipment and those who did not. The results showed that the average value of the questionnaires from the experimental condition (with VR equipment) was higher than those of the control condition (without VR equipment) on aspects of teaching equipment

Table 8. Two-Independent-Sample tests of social presence in the second lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
p24	20	26	29.25	760.50
	21	23	20.20	464.50
	Total	49		
p28	20	26	28.58	743.00
	21	23	20.96	482.00
	Total	49		
p32	20	26	28.63	744.50
	21	23	20.89	480.50
	Total	49		

Test Statistics[a]			
	p24	p28	p32
Mann-Whitney U	188.500	206.000	204.500
Wilcoxon W	464.500	482.000	480.500
Z	−2.284	−2.005	−2.015
Asymp. Sig. (2-tailed)	.022	.045	.044

[a]Grouping Variable : Group.

Table 9. Two-Independent-Samples tests of teaching and cognitive presence in the second lesson.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
r42	20	26	28.62	744.00
	21	23	20.91	481.00
	Total	49		
t45	20	26	29.75	773.50
	21	23	19.63	451.50
	Total	49		

Test Statistics[a]		
	r42	t45
Mann-Whitney U	205.000	175.500
Wilcoxon W	481.000	451.500
Z	−2.321	−2.692
Asymp. Sig. (2-tailed)	.020	.007

[a]Grouping Variable : Group.

perception (M: Z = −2.148, $P = 0.032 < 0.05$), class perception (N: Z = −2.345, $P = 0.019 < 0.05$), social presence (P: Z = −2.245, $P = 0.025 < 0.05$), and teaching presence (T: Z = −2.158, $P = 0.031 < 0.05$), as shown in Table 10.

Significant differences exist in the following nine items, as Table 11 shows. First, in terms of teaching equipment perception (m), there were significant differences between the experiment and control in "small words are clearly visible" (M1: Z = −2.123, $P = 0.034$) and "I feel it is as in life" (M7: Z = −2.816, $P = 0.005$). Second, in the aspect of class perception (n), there were significant differences in the three items of "I think learning content is rich" (N9: Z = −2.675, $P = 0.007$), "The presentation of learning content is easy to understand" (N13: Z = −2.567, $P = 0.010$), and "I feel the help from teacher" (N14: Z = −2.582, $P = 0.010$). Finally, significant differences exist in four items of social presence (p) including whether the learning environment was "personal" (P18: Z = −1.987, $P = 0.047$) and "appealing" (P24: Z = −2.582, $P = 0.010$), as shown in Table 11.

Table 10. Two-Independent-Samples tests of two lessons in two classes.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
M	0	47	51.71	2430.50
	1	44	39.90	1755.50
	Total	91		
N	0	47	52.24	2455.50
	1	44	39.33	1730.50
	Total	91		
P	0	47	52.01	2444.50
	1	44	39.58	1741.50
	Total	91		
R	0	47	51.74	2432.00
	1	44	39.86	1754.00
	Total	91		
T	0	47	50.86	2390.50
	1	44	40.81	1795.50
	Total	91		

Test Statistics[a]					
	M	N	*P*	R	T
Mann-Whitney U	765.500	740.500	751.500	764.000	805.500
Wilcoxon W	1755.500	1730.500	1741.500	1754.000	1795.500
Z	−2.148	−2.345	−2.245	−2.158	−1.832
Asymp. Sig. (2-tailed)	.032	.019	.025	.031	.067

[a]Grouping Variable : Group.

Table 11. Two-Independent-Samples tests of concrete items in two lessons in two classes.

Ranks				
	Group	*N*	Mean Rank	Sum of Ranks
m1	0	47	51.00	2397.00
	1	44	40.66	1789.00
	Total	91		
m7	0	47	52.47	2466.00
	1	44	39.09	1720.00
	Total	91		
n9	0	47	52.13	2450.00
	1	44	39.45	1736.00
	Total	91		
n13	0	47	52.15	2451.00
	1	44	39.43	1735.00
	Total	91		
n14	0	47	52.17	2452.00
	1	44	39.41	1734.00
	Total	91		
p18	0	47	50.57	2377.00
	1	43	39.95	1718.00
	Total	90		
p24	0	47	52.63	2473.50
	1	44	38.92	1712.50
	Total	91		

Test Statistics[a]	
	p24
Mann-Whitney U	722.500
Wilcoxon W	1712.500
Z	−2.582
Asymp. Sig. (2-tailed)	.010

[a]Grouping Variable : Group.

Surprisingly, VR's effect on teaching and learning is not as positive as expected because its educational value is closely related to the difficulty of teaching content. There is no significant difference between OHP and VR-assisted instruction in easy-to-understand content learning. When the learning content is difficult to understand, VR can promote student learning more than OHP can. However, on the whole, these results demonstrate that teaching with VR can enhance students' learning participation in the classroom. In term of teaching content, the first lesson in this experiment was relatively simple, and the second lesson was more difficult. In this study, the first lesson taught "appreciation of classical architecture" and students used VR devices to rotate and scale the details of buildings in order to observe them clearly; in this lesson, students mainly learned by themselves via observation. The second lesson taught "characteristics of the nervous system". The example of the second lesson suggests that using VR can help students to understand knowledge more comprehensively. While using VR in the first lesson only moderately enhanced students' social presence, the impact of using VR in the second lesson was more remarkable; these impacts were found in teaching equipment perception, classroom perception, teaching presence, and cognitive presence.

In summary, the results show that the educational value of VR is correlated with the difficulty of learning content. The impact of VR was more dramatically reflected in the second class lesson; this can be attributed to the difference between teaching in the experimental condition and the OHP condition, yet is also related to the difficulty of the teaching content. Compared with using VR in observational learning, using it in operational learning makes better use of VR's benefits, so students' perceptions and learning experience can be significantly improved.

Analysis of students' suggestions for using VR in two lessons

The above analysis verifies that VR significantly enhances the student's learning experience and has positive effects on the learner's perception and presence in secondary classroom teaching. Two open-ended items in the main questionnaire were designed to collect students' preferences and suggestions for using VR to teach, and obtain the objective evaluation of students on using VR assisted learning. The first question asked students about their preferences and viewpoints for VR assisted learning, and the second question asked about suggestions and strategies for improving VR application. Students' feedback about their preferences and reasons for VR-assisted learning in the two lessons mainly focused on the following points: the strong interaction, the strong three-dimensional sense, the freshness, and the relaxed and more active classes. Students reported showing more interest and care while using VR devices to learn. These answers also indicated that using VR to learn is more interactive and realistic. For the second question about suggestions and strategies for improving VR in the classroom, most students said that the equipment was very good, and they had no suggestions for improvement. However, a small number of students said that VR was tiring to the eyes and suggested that screen resolution, eye protection should continue to be improved.

Students' feedback on these two items shows that most students had a positive attitude toward VR-assisted learning. They thought that using VR made learning easier, and it relieved the learning pressure for some students. But a few students also pointed out that this teaching was too simple. VR-assisted learning, like all teaching media, cannot meet the needs of all students. Therefore, using VR-assisted learning requires teachers to teach students in accordance with students' aptitude and set tasks with different levels of difficulty for different levels of students.

Discussion

In order to dig out from the data that can inform teaching practice, unstructured interviews were conducted as a complementary on the practical significance of VR on teaching and other suggestions. The interview results of teacher A are presented as below. Teacher A reckoned that in current K-12 teaching environment, learning based on virtual reality(VR) often fails to achieve the set goals due to such influencing factors, such as equipments, teaching resources, and environmental construction foundation. The shortage of teachers with VR teaching experience and VR equipment have hindered current VR teaching in schools. Teacher A made two practical suggestions on how to use VR in teaching: Firstly, for designing and development of instructional resource, courseware for VR teaching need be designed with more focus on learner-center learning as well as inquiry skills. Secondly, VR teaching requires higher control and maintenance of environments. Several influencing factors are key points that equipping and selection of hardware, refinement and logical arrangement of teaching content during lesson preparing, and teacher's ICT skills. Only by careful preparation and well-design can the VR engaged lesson be carried out smoothly.

In addition, teacher A also made two suggestions for VR developers. Firstly, the developers could build a multi-channel environment by using VR scenes, pictures, audio and video to complement each other, thus providing students with a multi-sensory immersive learning environment. Secondly, some physiological responses of learners in moving and interaction can be collected, such as heart rate, respiration and pupil and so on. Therefore, learner's physical discomfort, fear or reluctance during the movement can be judged based on these physiological feedback cues. Then, the VR learning environment could give some encouragement and guidance, helping learners integrate into the learning context and allowing a embodied cognition. Learners would then be more engrossed in the learning in the virtual environment, producing a flow experience. They would not easily distracted by the outside world, and later through the contextual association of learning process, the learner's knowledge and memory would be stronger.

Overall, students' evaluation of classrooms using VR devices is slightly higher than classrooms without VR. Most student feedback consisted of positive evaluations of using VR in class. By communicating with teachers and students by Wechat (a communication tool). Reasons for students' positive evaluations include the following points: First, the VR devices are more popular with students. Students note that they have few chances to use these virtual devices in authentic learning in class. Novelty devices that one can interact with are quite different from traditional teaching and learning, for VR helps construct an independent virtual teaching observation space. Second, many teaching problems can be solved from the visualization of objects. Teachers using VR can present learning content in three dimensions, which makes it easier to understand unclear areas caused by two-dimensional pictures or videos. Through VR devices, students can zoom in to specific parts and observe carefully. Third, VR devices immerse students in learning content, which reduces the cognitive load of learning content as well as the stress of teaching pressure. In a VR-based class, classroom atmosphere is active and cheerful. Students' cognitive processes and cognitive pressure are less than the norm in a traditional classroom, leading to a pleasant learning process.

In addition, although VR equipment has greatly improved, and students generally have a positive attitude toward VR-assisted learning, there are still problems in the application, such as that students report that their eyes feel tired after using VR for a long time. Therefore, teachers need to organize teaching content and arrange learning interaction more carefully in a VR-based class. Teachers also need to provide more help to some students with learning difficulties in using VR devices, when necessary. Generally speaking, teachers should pay more attention to how to use VR to promote teaching interaction. Only when students take the initiative to participate in the classroom can the learning effect and results be really improved in a VR-based learning class.

Conclusion

This study explores the educational value of VR from the perspectives of how students view teacher's social presence based on empirical methods. The CoI framework is used in this study to investigate students' sense of presence and perception in real classroom teaching situations using VR. The study found that in specific instructional contexts, using VR can effectively enhance the student learning experience.. The effect highly depends on the matching of teaching materials and VR technology. using VR has better effects on transferring ill-structured knowledge than on transferring that of well-structured knowledge..

The findings of this study show that students recognize VR-assisted learning as beneficial to varying degrees with respect to perception of teaching equipment, perception in class, social presence, and teaching presence. However, students' evaluation of cognitive presence was not significant. The research data indicates that students in classrooms have similar performance in cognitive presence whether VR is used or not. In both of the two lessons studied here, students were highly satisfied with learning resources and teaching methods.

There are limitations to the present study that should be noted. The participants in this experiment were students from a high school in Beijing, China; Those students may have higher levels of achievements and technical skills than students in rural areas do. In the future, more participants, and more participants from more varied backgrounds, need to participate. Also, exploratory analysis involving more influencing factors needs to be conducted.

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